

# Anatomy and Biomechanics of the Hand

# 3

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Isabelle Schöffl bouldering at Halfway Log Dump, Ontario, Canada,  
Photo Michael Simon

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### 3.1 Anatomy

The flexor and extensor tendons must be looked at separately, and the lumbricalis muscles are yet another entity [1]. The complex movements of the fingers and wrist are only possible through a combined effort of the extrinsic and intrinsic muscles [2]. The extrinsic muscle system describes the muscles originating from the elbow and forearm and inserting into structures within the hand. The intrinsic muscles are situated completely within the hand. They are divided into four groups: the thenar, hypothenar, lumbrical, and interosseous muscles. In addition many anatomical variants exist [3].

#### 3.1.1 The Extensor Tendons

The long fingers have four common extensor tendons as well as two tendons which are dedicated to a single finger—extensor indicis for the second digit and extensor digiti minimi for the fifth digit. The tendon of the extensor digiti minimi runs through the fifth tendon compartment, while all other tendons run through the fourth compartment [1]. On the dorsum of the hand at the level of the metacarpophalangeal joints, there are many cross-connections known as the connexi intertendinei. At the proximal interphalangeal joint (PIP), the extensor tendons separate into two lateral reins and one central rein (tractus intermedius). The thumb has two extensor tendons, the extensor pollicis longus (third extensor compartment) and the extensor pollicis brevis (first extensor compartment). Together, these tendons form the so-called extrinsic system, tendons of muscles which originate proximal of the hand itself [1] but are moving the hand and fingers. The extrinsic system is supported by the intrinsic system, muscles originating within in the hand, the mm. lumbricales, the mm. interossei, and the thenar and hypothenar muscles [1]. The lumbricalis and interossei are exceptions as they originate from the flexor tendons themselves and end in the tendinous hood of the extensor tendons [4]. Their function is flexion in the metacarpophalangeal joint (MCP) and extension in the proximal

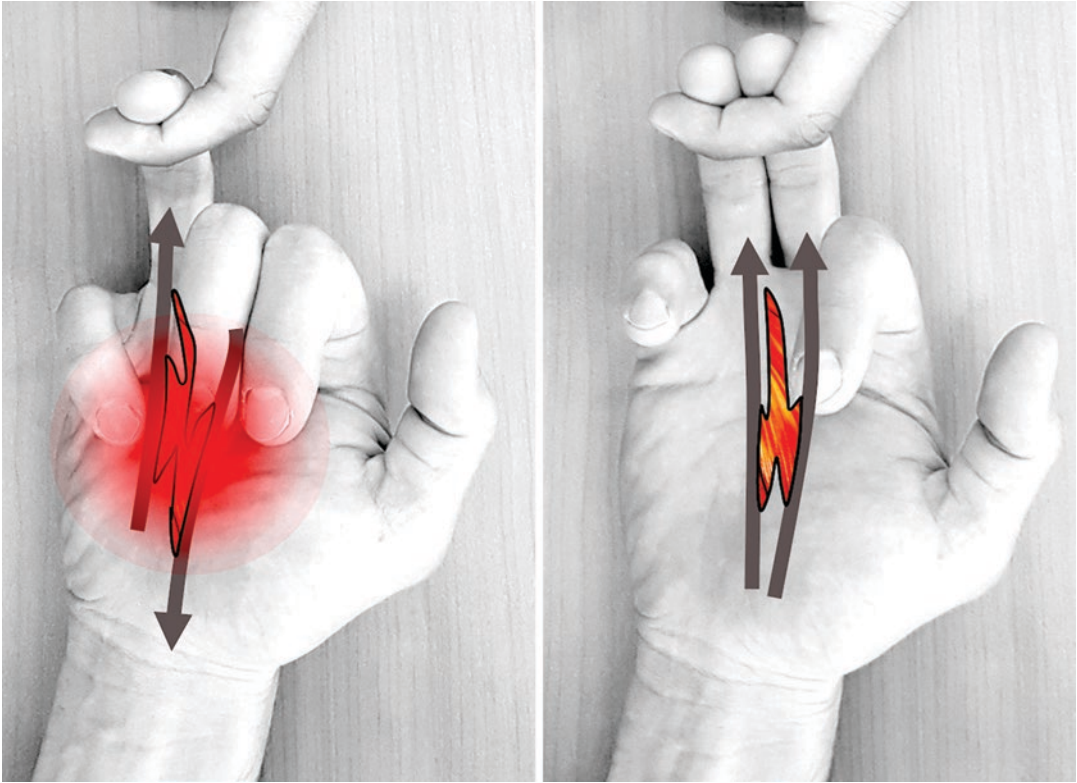
(PIP) and distal interphalangeal joint (DIP) [1, 4, 5]. The muscle bodies of the lumbrical muscles have many anatomical variants and often have one common muscle body for the muscles of two adjoining fingers, mostly for the fourth and fifth finger [3, 4, 6]. If one of these fingers is extended and the adjoining one is flexed, the common muscle body receives shear force, which was reported by Schweizer as “quadriga” effect and leads to a “lumbrical shift” phenomenon [6] (Fig. 3.1).

#### 3.1.2 The Flexor Tendons and Their Functional System with the Pulleys and Tendon Sheath

The flexor tendon apparatus has to be considered as a functional unit of tendons, tendon sheath, and pulley system [1]. The long fingers and the thumb differ with respect to the flexor tendons. The long fingers have two flexor tendons, the flexor digitorum profundus (FDP) and flexor digitorum superficialis (FDS) which run through the carpal tunnel and pulleys and intersect at the chiasm [1].

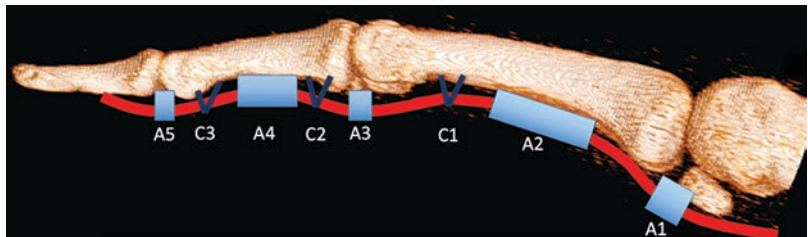
The one thumb flexor tendon (flexor pollicis longus) runs through the carpal tunnel, on the radial aspect of the forearm and then through an osteofibrous channel up to the base of the phalanx and is reinforced by two pulleys [1].

The annular ligaments and the cruciate ligaments represent a reinforcement of the tendon sheaths. Thus, the tendons are guided in an osteofibrous channel to the phalanges [7]. Five annular (A1–A5) and 3 weaker crucial ligaments (C1–C3) can be distinguished [7, 8] (Fig. 3.2). Pattern and arrangement of these ligaments vary with a high diversity for the A3 and C1/C2 pulleys [8, 9]. All pulleys have different functions in stabilizing the flexor tendons at the palmar sides of the phalanges [8, 10–12]. The main function of the flexor tendon pulley system is to hold the flexor tendons close to the bone, thus converting linear force into torque resulting in rotation at the interphalangeal (IP) and metacarpophalangeal (MCP) joints [13]. The A2 and A4 annular ligaments play the most important role in force transmis-



**Fig. 3.1** The “quadrige” effect of the lumbrical muscles [6]

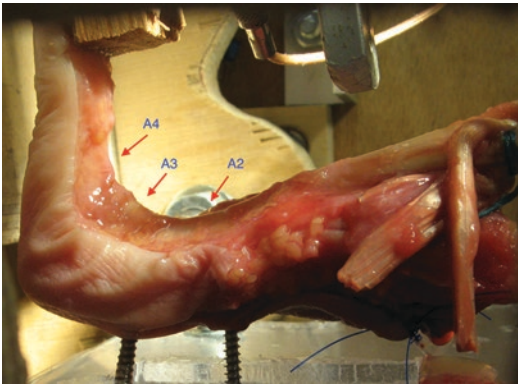
**Fig. 3.2** The finger flexor tendons pulley system



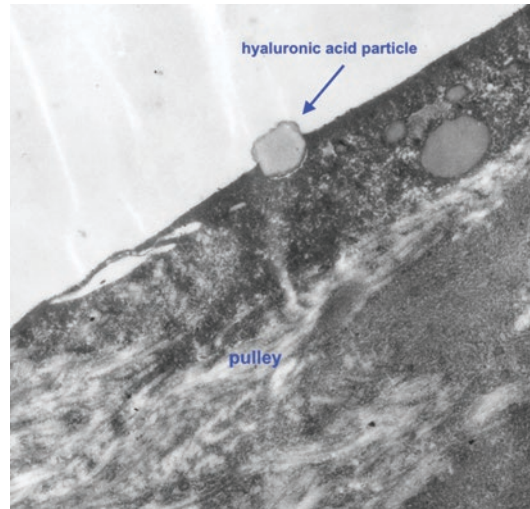
sion and deflection of the flexor tendons [14–17]. A minor role is performed by the A1 and A5 pulleys, which are expandable [13]. However, the A3 pulley also controls the extent of tendon bowstringing as it is situated at the PIP joint and is thus more effective in maintaining the tendons close to the phalanges [18–21].

The pulley system keeps the tendons close to the bone. The force of the flexor tendons is transferred efficiently in flexion and hyperextension to reach the full range of motion [13]. A loss of one or several of the pulleys will cause an increased distance between the flexor tendons and the bone

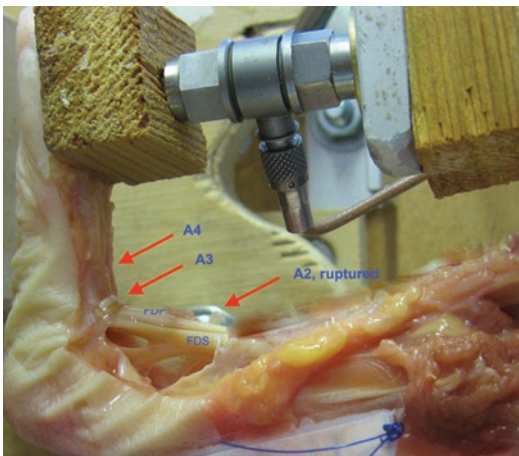
(tb = tendon-bone distance), which is called “bowstringing” and leads to a loss of strength and a decreased range of motion [7] (Figs. 3.3 and 3.4). These deficits are a function of the specific pulley or pulleys being ruptured. The bowstringing is minor, if only one pulley is ruptured and then can only be detected via ultrasound or MRI/CT imaging. If more than one pulley is ruptured, the bowstringing is visible and palpable and also leads to a decreased range of motion in the PIP joint. The A2 and A4 pulleys are believed to be the most important ones in preventing bowstringing and thus ensuring optimal force transmission [12, 18, 22,



**Fig. 3.3** Complete pulley system in a cadaver stress test in the biomechanical lab



**Fig. 3.5** Electron microscopy of a pulley which secretes hyaluronic acid [1]



**Fig. 3.4** Torn A2 pulley in a cadaver stress test in the biomechanical lab. The increased distance of the flexor tendons to the bone is clearly visible, the A3 and A4 pulley are intact

[23]. Over time the increased tendon bone distance after multiple pulley ruptures performs as an increased lever arm onto the center of rotation of the PIP joint, causing contracture. In addition to their mechanical function, the pulleys themselves are also important for nutrition of the flexor tendons and actively secrete hyaluronic acid into the tendon sheath [10, 11] (Fig. 3.5).

The lumbricalis and interossei muscles are exceptions as they originate from the flexor tendons themselves and end in the tendinous hood of the extensor tendons [4]. They are part of the intrinsic muscle system. Their function is flexion

in the MCP and extension in the proximal (PIP) and distal interphalangeal joint (DIP) [1, 4, 5].

Blood supply of the flexor tendons is guaranteed through the “vinculae tendinae” in the region of the osseous insertion of the tendon as well as in the osteofibrous channel [1, 4, 5]. Venous drainage is performed through the same system [1]. Verdan [24] divided the flexor and extensor tendons into different regions of interest regarding injuries, prognosis, and nutrition [25].

### 3.1.3 The Physis (Growth Plate)

The physis is situated between the epiphysis and the metaphysis of long bones. It enables the bone to grow and is only detectable in children and adolescents. When growth ceases after the end of puberty, the physes become fully calcified and cannot be detected any longer. The physis is the place of endochondral calcification. It is made up of chondrocytes, at different stages of differentiation [26]. There are basically three different stages of different differentiation, termed zones. The chondrocytes in the resting zone replicate at a slow rate and replenish the pool of proliferative chondrocytes [27]. The chondrocytes in the proliferative zone replicate at a high rate resulting in cells lining up along the long axis of the bone

[26]. Once the cells stop dividing, they differentiate into hypertrophic chondrocytes increasing their height six- to tenfold, making up the main part of longitudinal growth [28]. The hypertrophic chondrocytes then calcify the surrounding extracellular matrix and produce factors that attract the invading bone cells and blood vessels, including vascular endothelial growth factor [29], before undergoing apoptosis.

## 3.2 Tendon Healing

Histologically, tendons consist of extremely long collagenous fibers which are arranged into bundles. Similar to a rope ladder, elastic fibers and vessels are entangled in between those bundles [30]. To provide low friction during movement, tendons are covered by a paratendineum or a tendon sheath. Pulleys strengthen these sheaths along the phalanges. Friction plays an important role in the origin of injuries and chronic inflammatory diseases of the tendons, their sheaths and pulleys [10, 11, 31, 32]. Following an injury, healing processes emerge from the paratendineum and the paratendinous tissue, necessitating a distinction of an extrinsic and an intrinsic healing process [5, 30, 32, 33]. Characteristic for an extrinsic healing process is a distinctive inflammatory response followed by proliferation and remodeling [1]. Fibroblasts of the paratenon play an important role in migration which leads to adhesions. Immobilization supports the adhesion process [32–36]. The intrinsic healing process, supported by movement of the tendon, is characterized by immigration of “fibroblast like tenocytes,” which produce the collagenous tissue and carry out the remodeling process [30, 32–36]. The inflammatory response is minimal, the clinical outcome better. This is the ratio of the widely recommended early passive movement therapy, which leads to a better nutrition and strength of the tendon [30, 32–36]. The following factors predict the tendon healing: age, overall health, scar formation disposition, motivation, injury risk based on Verdan’s zones [24], injury type, synovial containment, as well as the surgical technique [1, 32]. Three phases of tendon healing

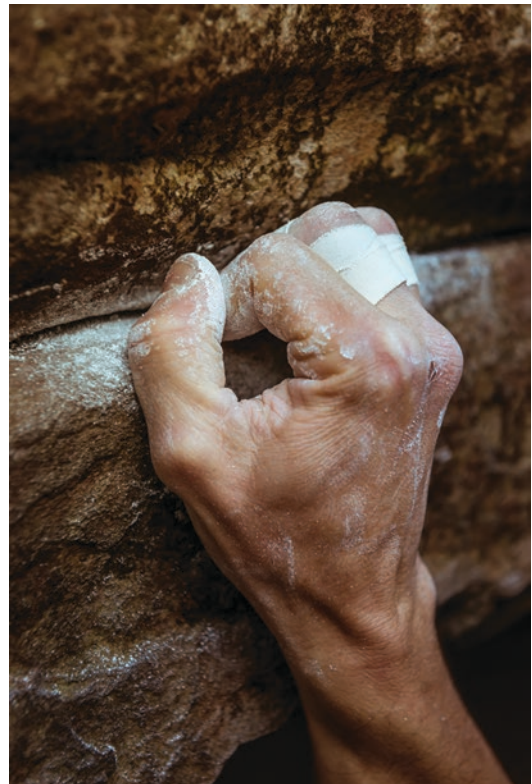
are defined. First, a migration of peripheral cells and invasion of blood vessels occur; second, the tendon and surrounding tissues heal. Remodeling happens in the third phase of healing due to movement and function of the tendon [37].

The tendon gains its daily life-loading capacity after 12 weeks of healing; sportive activities are allowed earliest 4 months past injury. The remodeling process can last up to 12 months [1].

## 3.3 Biomechanics

### 3.3.1 Pulleys

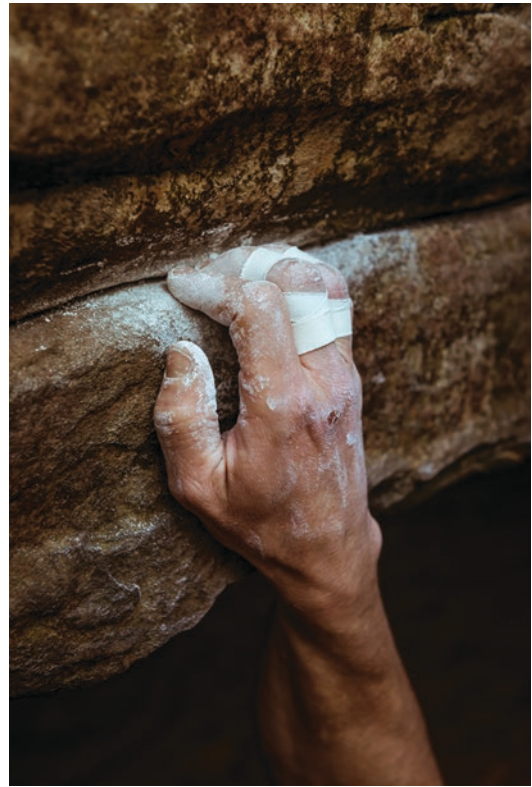
In climbing two main finger positions are used for holding small edges, the hanging and the crimping position (Figs. 3.6 and 3.7). Additionally, many other hand positions exist for the various other shapes of holds, be it indoors or



**Fig. 3.6** The crimping position transfers a high stress load onto the finger flexor tendon pulleys and onto the joint cartilage. (Photo by Kilian Reil)



**Fig. 3.7** In the hanging finger position, the passive structures, the joint capsule, the flexor tendons, and the ligaments receive the highest load. (Photo by Kilian Reil)



**Fig. 3.8** The “half-crimp” reduces the stress onto the fingers. (Photo by Kilian Reil)

outdoors. Finger locks in crack climbing, underclinging, one- or two-finger pockets, pinch grip, stacked fingers grip, etc. are also frequently used hand and finger positions. Lately the “half-crimp” (Fig. 3.8) became popular as it is apparently placing less stress onto the pulleys and cartilage of the finger joints [3, 13].

However, the crimp grip and the hanging finger position are the most common especially for holding on to small edges. In the crimp position, the distal interphalangeal joints are hyper-extended, the proximal interphalangeal joints flexed, and the metacarpophalangeal joints extended. The wrist joint is extended to increase strength development for the finger flexion [38]. In the hanging finger position, the PIP, DIP, and MCP joints are flexed, while the wrist can be in any position from flexed, neutral, to extended. The crimping position helps

to increase strength transfer in holding very small edges; thus, many climbers use it for these kinds of holds [39]. The crimping position leads to high pressure on the cartilage of the small finger joints and high stress on the pulley system. In the hanging position, most stress is transferred to the passive structures, the tendons, joint capsule, and its ligaments. The half-crimping position has recently been recommended more and more often as it decreases the high stress of the full crimp, while it still allows good strength [40]. Two other finger positions lead to biomechanically based stress, the “stacked” finger position (Fig. 3.9), and the “mono” (one finger pocket, additionally also two finger pocket) (Fig. 3.10). In the stacked finger position, a shear stress is forced onto the MCP and PIP joint as well as to the connexus intertendineus of the extensor tendons. The mono (one finger



**Fig. 3.9** A stacked finger position leads to shear forces onto the MCP and PIP joint as well as the connexus intertendineus



**Fig. 3.10** The one finger pocket (mono) leads to a high stress onto the joint capsule and the flexor tendons and due to the fact that the adjoint fingers are flexed to a “quadriga” effect of the lumbrical muscles (Photo by Michael Simon)



**Fig. 3.11** Clinical bowstringing in a A2/3/4 rupture

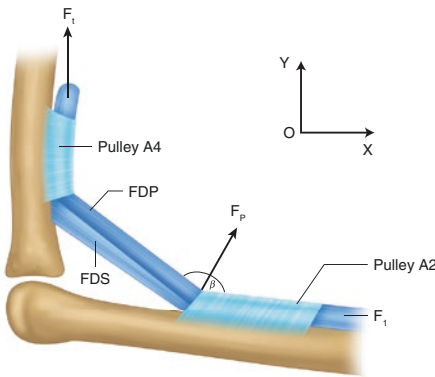
pocket) and the two finger pocket positions lead to a high stress onto the joint capsule and the flexor tendons and due to the fact that the adjoint fingers are flexed, to develop more strength, to a “quadriga” effect of the lumbrical muscles [6, 40].

The main function of the flexor tendon pulley system is to hold the flexor tendons close to the bone, thus converting linear force into torque resulting in rotation at the interphalangeal and metacarpophalangeal joints [13]. A loss of one or several of the pulleys will cause an increased distance between the flexor tendons and the bone, a “bowstringing,” and leads to a loss of strength and a decreased range of motion [7, 18]. Figure 3.11 shows a clinical bowstringing and Fig. 3.12 a cadaver specimen after iatrogenic A2/3/4 rupture in our biomechanical lab. The clinical picture of a loss of function is depending on the respective injured pulley. The A2 and the A4 pulley are considered as the most important pulleys to prevent a bowstringing, while the A3 pulley is weaker and more flexible, thus having a minor impact onto the flexor tendons course. It also does not insert onto the bone but on the palmar plate.

The forces acting on the pulleys during climbing can be estimated using mathematical models which incorporate anatomical observations into mathematical formulas [13, 41] (Fig. 3.13). The pulley deflects the tendon at the point of contact, and thus



**Fig. 3.12** Bowstringing in a cadaver finger in the biomechanical lab



**Fig. 3.13** Important for the force onto the A2 pulley ( $F_p$ ) is the angle  $\beta$ , which depends from the flexor angle of the PIP joint [13]

the force acting on the pulley ( $F_p$ ) is a function of the angle between the tendon and the pulley ( $\beta$ ) and the force developed in the tendon ( $F_T$ ):

$$F_p = 2F_T \cos\left(\frac{\beta}{2}\right)$$

Furthermore, there is a relative mismatch between the width of the phalanx and the width of the tendons which leads to differing forces according to this mismatch.

$$F_i = F_p \left( 1 + \left( \frac{L_B - L_T}{2 * H_T} \right)^2 \right)^{\frac{1}{2}}$$

$F_i$  represents the force acting on the pulley insertion,  $L_B$  and  $L_T$  the width of the bone and tendon, and  $H_T$  the height of the tendon.

And finally, pulleys have a certain potential to stretch when loaded so that the angle between the tendon and the pulley will decrease once the pulley is loaded. Therefore, the stiffness of the pulley needs to be considered when modeling the forces acting on the pulley during climbing [15].

Simplified, it can be concluded that once the finger is flexed, the angle between the pulley and the tendon decreases, and the force acting on the pulley increases. This in turn signifies that a crimp grip position leads to higher forces acting on the pulley system than a hanging finger position. The higher the mismatch between the tendon and the bone width is, the higher the forces acting on the pulley.

Consequently, a maximum flexion in the PIP joint leads most frequently to a pulley rupture. These high forces are obviously happening during climbing but can also happen during other activities while stressing the fingers in a crimp grip position (e.g., lifting heavy items) [42]. The pulleys which are located around the PIP joint (A2, 3 and A4) are the ones which are mostly affected [31], but also singular c pulley injuries are reported [43].

The smaller this angle becomes, the higher the force is onto the respective pulley [13].

Butcher and Jenny [44] mathematically analyzed forces in various finger positions and found a force of 599 N on the pulley system with a PIP joint flexion of 135°. Biomechanical analyses of Bollen [22] estimated forces up to 450 N acting on the A2 pulley, and a model by Roloff et al. [13] calculated forces of 268 N for the A4 pulley in the crimp position, to be 268.8 N. Considering these high forces acting on the pulley system, the maximal tear load of pulleys needs to be considered. Lin et al. [45] found a maximal tear load of the A2 pulley of 400 N, while Widstrom et al. [46, 47] found 137 N. Even though these measurements show a wide variety, these results show that the forces estimated to act on the pulleys in the crimping position are very close to their biological tearing force in vitro. If, in addition to these high forces, the climbers hand slips off a hold or a foot comes loose, the biological strength of the pulley system is bound to fail.

Most cadaver studies on pulley failures used concentric forces onto these pulleys, mostly in

behalf of pulling onto the pulleys until they tore [10]. Nevertheless, this may not be the real stress leading to rupture. Most injured climbers report that the movement leading to their pulley rupture can be described as an eccentric movement: passive opening of the hand, slipping off a foothold with a sudden increase of force on the pulley system, or a strength deficit while trying to hold onto a hold [39, 48, 49]. The analysis of a normal movement pattern in climbing shows that there is a pulling, concentric movement of the arm, with flexion of the elbow, but higher up the chain, there is an eccentric movement of the PIP joint; the fingers are slightly opening up. Many climbers report that the injury occurred at the end of a longer climbing day, when trying to hold an edge in a tired state with the consequence of not being able to maintain the crimping position. This in turn leads to an opening of the fingers [7]. In further cadaver studies, we therefore examined pulley ruptures during concentric versus eccentric movements, as well as in hanging versus crimping position [10, 11]. We found that pulley ruptures happened more easily during an eccentric movement than during a concentric movement (failure load: eccentric movement, 237 N on FDP, concentric movement 354 N on FDP) [10]. The reason for this discrepancy between concentric and eccentric loading is probably due to friction between tendon and tendon sheath/pulleys. Studies on birds and bats have shown that the inner coating of their pulleys, facing the flexor tendons, has an explicit rough surface. As a consequence of the high friction in between this rough inner surface and the flexor tendons, these animals can hang passively on their claws, without active muscular activation. Similar mechanisms could be shown for birds [50].

Examinations of human pulleys via electron microscopy show similar patterns in humans. Schweizer [31] was able to measure this increased friction in human pulleys and their possible correlation to pulley rupture in an *in vivo* study. In conclusion, it can be said that friction underneath the pulleys is an advantage for a climber, as this will increase flexor strength. However, an increased friction between the inner surface of

the pulley and the tendons may be the reason why some climbers are more prone to pulley injuries than others. The obvious downside in “rough” pulleys and great finger strength is the higher incidence of pulley injuries in these people.

Even if the mathematical determined force values as well as the pulley tear forces in the cadaver models can only be partially transferred into the dynamic pulley flexor tendon system *in vivo*, these findings clearly show that in a crimping position the forces acting on the flexor pulleys are close to their tearing strength [7]. This explains the fact that a pulley rupture is one of the most common climbing-specific diagnoses. These high forces which are certainly occurring in climbing can also happen in daily live activities, e.g., lifting a heavy item [42]. The youngest patient with a pulley lesion was a 2.5-year-old girl, injuring her finger while playing in a sand box and her brother stepping onto the finger, while she was pulling it away.

The ring finger is most often concerned in pulley ruptures followed by the middle finger and the other fingers [7], which can possibly also be explained through a biomechanical analysis of the movement pattern.

In the crimp position, the carpal joint performs in addition to an overextension an ulnar abduction [7]. Caused by this ulnar abduction in combination with a mild supination while crimping, the ring finger receives the highest stress [51]. Especially as the ring finger is less powerful than the middle and the index finger, this rotational moment increases stress and injury potential to its pulleys [52]. Furthermore, the adjoint little finger is in most cases much shorter than the ring finger and therefore offers little support to the ring finger. Injuries to the index finger are rare, probably due to its decreased stress during the abovementioned supination. In crimping with the full hand, the middle finger is supported on both sides by the ring and index finger. The index finger is supported by the thumb which is often stacked on top of it, while nothing supports the ring finger. Pulley injuries to the middle finger are most frequent if the crimping is only performed with two fingers [53].

### 3.3.2 The Growth Plate

Fractures to the growth plate have previously been described in other sports on various anatomical locations. The first cases were observed in the proximal humeral growth plate of young baseball players and have been termed the Little Leaguer's shoulder [54–56]. The cause for this injury is believed to be a consequence of the whiplash action of the arm during a throwing movement [57]. As a consequence of repetitious throwing, a widening of the epiphyseal plate as in a Salter Harris I fracture [57] occurs. Therefore, most of the epiphyseal fractures observed in baseball are Salter Harris I fractures [55, 56]. After this first description of disruptions to the growth plate as chronic injury in baseball, stress fractures of the distal radius in gymnasts were being recognized on a regular basis [58–60]. The pathomechanism is believed to be complex rotational vaults [58, 61]. As a consequence of this loading mechanism, the injuries most often observed are Salter Harris II stress fractures with an epiphyseal widening and irregularity and cystic changes of the metaphyseal aspect of the growth plate [58, 59, 61]. The climbing community only started reporting epiphyseal stress fractures in the fingers in 1997 [62] and 1999 [63]. As a consequence of repetitive loading of the fingers, the fractures observed were always in the proximal interphalangeal joint [62]. Most often they were fractures of the Salter Harris III type with a fracture through the epiphysis of the middle phalanx [64].

All these injuries have repetitive loading as a common denominator. As there is no true blood supply to the physis but only an advance from the blood vessels of the epiphysis and metaphysis as well as from the perichondrial ring and vessels of the periosteum [65], repetitive loading can alter the metaphyseal perfusion and thus interfere with the mineralization of the hypertrophied chondrocytes [66]. The hypertrophic zone then widens as a consequence of constant growth in the germinal and proliferative zones. Usually, this widening of the epiphyseal plate is only temporary, as the resting and dividing cellular layer of the growth plate, and the attendant epiphyseal and metaphyseal blood supplies are essentially undisturbed. If

the ischemic condition sets in as a consequence of repetitive loading with too much weight, an osseous necrosis and deformity within the developing ossification center can be caused, leading to growth irregularities in the physis. As these changes localized, asymmetric growth can be observed, or the entire physis may be involved leading to an overall slowdown of the rate of growth or even complete cessation of growth in that joint [67].

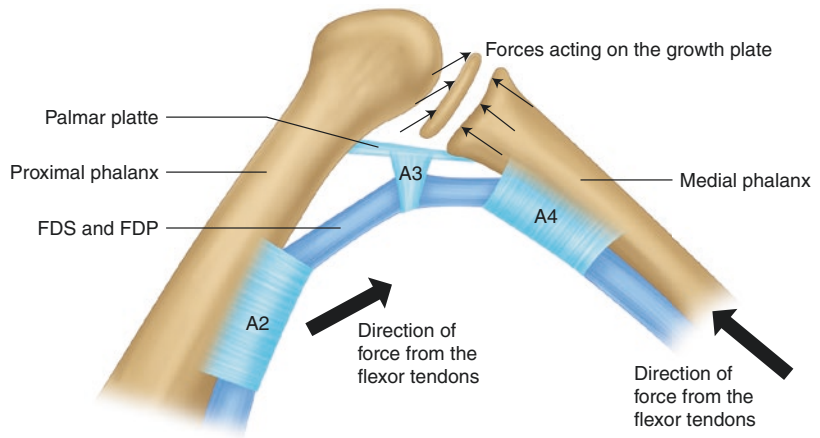
The growth plate seems to be exceptionally susceptible to injuries during periods of rapid growth [68–71] that is during the growth spurt during puberty. The strength of the physal cartilage has been shown to decrease during pubescence in animal studies [72] as well as in humans [73]. This decrease in strength is a consequence of the structural changes during rapid growth leading to a thicker and more fragile plate as well as to the bone mineralization lagging behind the linear growth, rendering the bone more porous [74]. Furthermore, shear resistance is a function of the amount of matrix and collagen. Increased cell size in the hypertrophic zone reduces the amount of matrix available to resist shear, and an increase in weakness is thus predictable [75].

The process of longitudinal bone growth is governed by a complex network of endocrine signals, including growth hormone, insulin-like growth factor I, glucocorticoid, thyroid hormone, estrogen, androgen, vitamin D, and leptin [26]. However, how much the different hormone levels impact on the risk of growth plate fractures is questionable and has so far has not been examined.

Whether an increase in muscle-tendon tightness about the joints as a consequence to the growth spurt leads to an excessive muscular stress on the physis is controversial in the literature [71, 76, 77].

The mechanism behind this injury is believed to be a high repetitive stress on the proximal interphalangeal joint using a crimp grip position of the hand (Fig. 3.6) [64, 78]. In this finger position, the physis is subjected to a compressing force coming from the fingertip on one hand and from the flexor muscles on the other hand. As the finger is flexed to a maximum, this compressive

**Fig. 3.14** Forces acting on the growth plate of the finger, in the crimp grip position, with a resulting compressing force on the growth plate coming from the fingertip on one side and from the flexor muscles on the other side



force is uneven, applying more force to the dorsal aspect of the growth plate (Fig. 3.14). This in turn leads to the fracture line going through the dorsal aspect of the growth plate as well as the epiphysis. This is exaggerated in campus board training in which the athlete performs climbing movements with the feet off the ground, thus subjecting the fingers to the full body weight. A study investigating radiographic changes in the fingers of top level climbers revealed that only those climbers who performed this kind of training had osteoarthrotic changes [79, 80].

### 3.4 Conclusion

The so-called crimp grip position, which is commonly used in climbers, is a very specific position of the fingers where the proximal interphalangeal joint is flexed more than  $90^\circ$ , and the distal interphalangeal joint is hyper-extended, resulting in very high loads onto the pulleys of the flexor tendon sheath, which are three to four times higher than the forces acting at the fingertip [81]. Also, the existence of friction between the flexor tendons and the pulleys [10, 31] adds to the understanding of the pathomechanism of the rupture of a pulley, which mostly happens during eccentric movement [10]. Adolescents during their growth spurt are most prone to injure the growth plate due to forces pushing the physis out on the dorsal aspect of the PIP joint. An injury to

the growth plate can lead to serious impairment of the finger with malaligned growth or cessation of growth altogether. Campus boarding and the crimp grip position are the main causes for injuries to the growth plate.

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